10.10 Fuzzy queries

We can ask Elasticsearch to forgive spelling mistakes using fuzzy queries with `query_string` queries. All we need to do is suffix the query criteria with a tilde (~) operator. This is best understood with an example.

Listing 10.27 A fuzzy query string query

```
GET books/_search
{
  "query": {
    "query_string": {
      "query": "Pattenrs~",
      "default_field": "title"
    }
  }
}
```

Incorrectly spelled "Pattenrs" as the search word

By setting the suffix with the ~ operator, we ask the engine to consider the query fuzzy. By default, an edit distance of 2 is used when working with fuzzy queries. The *edit distance* is the number of mutations required to transform a string into another string. For example, the words *cat* and *cap* only differ by a single letter. Hence, we require an edit distance of 1 to transform *cat* into *cap*.

In chapter 9, we learned about fuzzy queries in the term-level context, which utilize the Levenshtein distance algorithm. Another type of edit distance algorithm is the Damerau–Levenshtein distance algorithm, which is used to support fuzzy queries in the full-text context. It supports insertions, deletions, or substitutions of a maximum of two characters as well as transposition of adjacent characters.

> **NOTE** The Levenshtein distance algorithm defines the minimum number of mutations required for a string to be transformed into another string. These mutations include insertions, deletions, and substitutions. The Damerau–Levenshtein distance algorithm goes one step further. In addition to having all the mutations defined by Levenshtein, the Damerau-Levenshtein algorithm considers transposition of adjacent characters (for example, *TB* > *BT* > *BAT*).

By default, the edit distance in a `query_string` query is 2, but we can reduce it if needed by setting it to `1` after the tilde: `"Pattenrs~1"`. In the next two sections, we look at some simpler queries.

10.11 Simple string queries

The `query_string` query is strict on syntax, and errors in the input are not forgiven. For example, the following query throws an error because the input has parsing problems (on purpose—we added a quote to the input criterion).